

A
Research Intern project
Report
On
Heat Action Economics
At



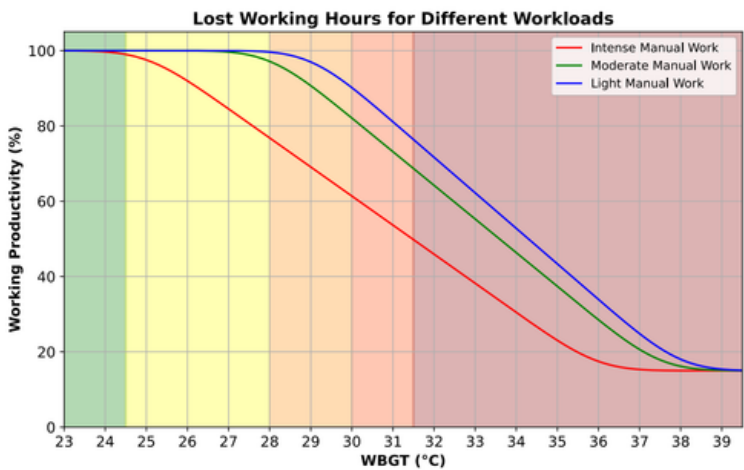
Guided By:

Prof. Disha Sharma
Assistant Professor
Sustainability Management and Centre of Policy
and Governance
M : +91 9999512244
Email: disha.sharma@iimmumbai.ac.in

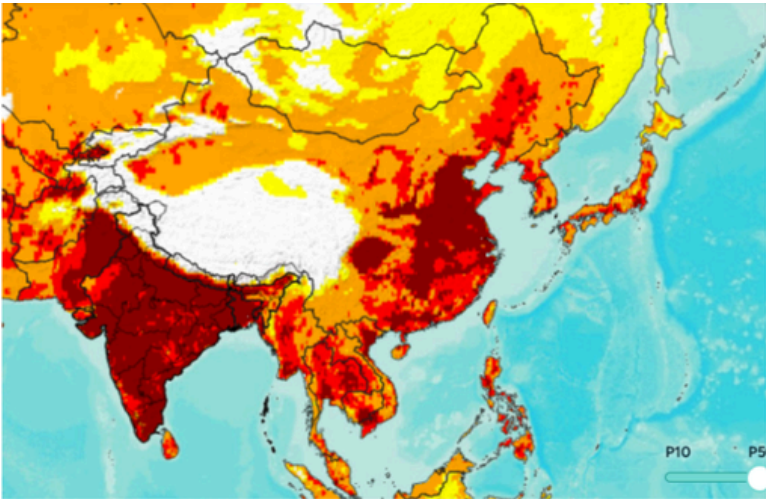
Economics of Extreme Heat: climate changes,Elnino,heat action plans and actions from India and China

Context

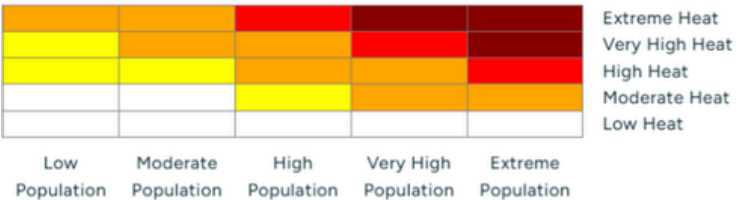
Heatwaves are a growing concern globally due to their adverse effects on nearly all aspects of urban life, including health, the economy and infrastructure. Extreme periods of hot temperatures are associated with increased deaths and illnesses, reduced productivity within heat-exposed sectors of the workforce, increased suicide and accident rates , higher operating costs and reduced asset lifespans for energy and transport infrastructure and reduced learning outcomes for schoolchildren



Graph depicting Loss of work capacity at varying heat stress levels for occupations of light, moderate and high physical intensity.



RISK FACTOR CATEGORIZATION



Graph depicting a heat-risk categorization map based on temperature and population exposure, showing India in the **extreme heat-risk category**

By 2030, extreme heat could melt away 4.5% of India's GDP. The World Bank report suggests there's a direct 150 Billion \$ direct threat to the economy. The reports predicts a loss of 4.5% of Indias GDP due to heat related labour productivity decline.Extreme heat and humidity are killing labor hours.**We lack critical "cold-chain" transport.** Without it, post harvest crops rot, rural incomes tank, and life saving vaccines spoil.

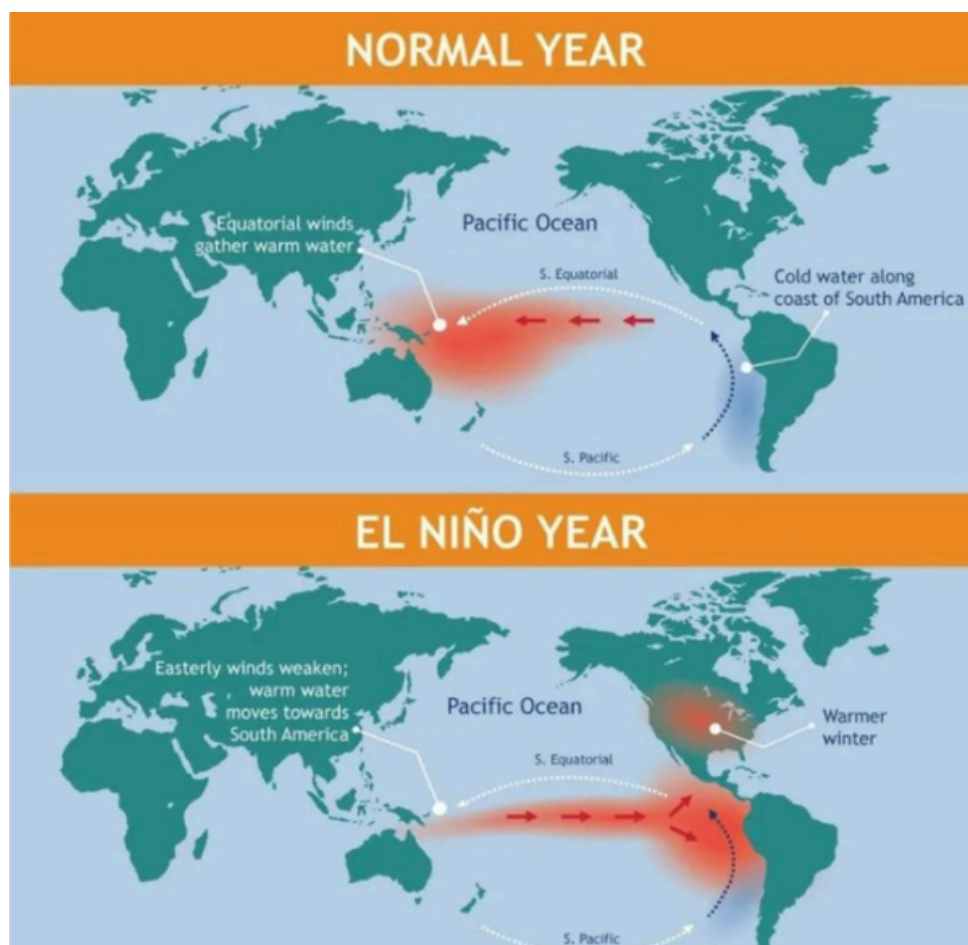
The fix isn't just installing more ACs. India needs "green cooling"; smarter building designs that naturally deflect heat and sustainable logistics.

UC berley, India Energy and Climate Centre (IECC) reports suggests,India's power grid recently hit a record-breaking peak demand of 270 GW, and room air conditioners (ACs) alone were responsible for a staggering 25% (60–70 GW) of that total.

ndia is only at the beginning of this cooling surge. Over the next ten years, ACs are projected to add another ~180 GW to the country's peak demand ,an amount larger than the entire peak power demand of Japan. we cannot treat this only as a supply side problem. Building more generation capacity is important, but it is not enough. We also need to actively manage cooling demand.

'The upcoming El Nino event could cost India \$1 trillion in losses'

El Niño emerges in Pacific, raising heat risks and crop threats



Picture comparing normal Pacific Ocean climate pattern with an El Niño year, showing that during El Niño, warm ocean water shifts east toward South America, weakening trade winds and altering global weather patterns.

It's still May and the year is turning out to be a case of when it rains, it pours.

This year, it looks like a super El Niño is developing, and the IMD is already forecasting **rainfall 6% below** normal for 2026.

It may not sound like much, but remember, 70% of India's annual rainfall comes from the monsoon, and 60% of farmers depend entirely on it. If history is any guide, we may have a terrible year ahead. In 60% of El Niño years since 1951, India has seen below-average rain. In 2009, rainfall fell to just 78% of normal, the worst in 37 years.

A weak monsoon means weaker harvests, and weaker harvests mean higher food prices and higher inflation. Food is one of the biggest expenses in a household budget. .

US Iran conflict has effectively shut a channel that carries 20% of the world's oil and 20% of its LNG. with price of crude oil reaching approx 100\$ per barrel and India importing 80 to 90% of its oil and 40 to 50% of its gas, we are already seeing steady price hikes and WFH advisories going out around the world. When food and energy prices rise together, the RBI cannot stay quiet. Beyond a point, it will have to start hiking rates, and that is when a bad situation starts to feel like a crisis.

Economics Times article covering interview of **Associate professor of geography** says On basis of current forecasts results, upcoming El Niño event, expected to peak between November 2026 and January 2027, may slow India's economic growth enough to cause cumulative losses exceeding \$1 trillion by 2032, representing economic output that would

have been generated under normal climate conditions. While India's economy is still expected to grow, the growth rate could be noticeably lower because of disruptions caused by El Niño, such as impacts on agriculture, labor productivity, energy demand, and public health. On a global scale, the total economic cost of this event could surpass \$10 trillion, highlighting the substantial long term economic risks associated with extreme climate phenomena.

Estimating economic loss due to heatwave days leading to abrupt rainfall due to southwest monsoon winds

We used the concept of multiple linear regression. The aim is clear,Heatwave events causes abrupt rainfall days all of which directly impairs crop cultivation and thus leading to loss For reference, We took the Punjab state. Gathered the relevant data, performed our analysis on MATLAB simulation and this is what we got.

FORMAL SPECIFICATION

$$WheatYield_t = \beta_0 + \beta_1Temperature_t + \beta_2Rainfall_t + \varepsilon_t$$

where

WheatYield_t = Punjab wheat yield (kg/hectare) in year *t*

Temperature_t = Annual mean temperature (°C) in Punjab

Rainfall_t = Monsoon rainfall departure (%) or monsoon rainfall (mm)

β₀ = Intercept (baseline wheat yield)

β₁ = Effect of temperature on wheat yield

β₂ = Effect of rainfall on wheat yield

ε_t = Other factors not included in the model

the following table of annual observations,spanning 10 years(2015 to 2024) was used for the iterations

A	B	C	D
Year	Wheat yield(kg/ha)	Heatwave days	Rainfall(mm)
2015	4294	3	546
2016	4583	5	427
2017	4704	12	493
2018	5077	4	598.3
2019	5188	8	578
2020	5033	1	450
2021	4868	2	530
2022	4213	24	414
2023	4748	5	650
2024	5045	22	391

the following table of annual observations,spanning 10 years(2015 to 2024) was used for the iterations

Running the code on the MATLAB Simulation resulted us with the following output and economic loss for **4 million hc of cultivatable land** for a **MSP of ₹ 2275** as per 2024-25 data

```
Intercept(b0)= 4580.9591
Heatwave Coef(b1)= -6.8715
Rainfall Coef(b2)= 0.4992
R^2= 0.0708
Adj R^2=-0.1947
Yield Loss=0.0687 quintal/ha
Total Yield Loss= 0.275 million quintals
Economic Loss= Rs 0.625 billion
```

Results indicate that an **additional heatwave day is associated with a reduction of approximately 6.87 kg/ha in wheat yield**, while higher rainfall has a small positive effect. The estimated yield loss attributable to heat stress is 0.0687 quintals per hectare, translating into a total production loss of 0.275 million quintals and an economic loss of approximately ₹0.625 billion (₹62.5 crore).

HEAT ACTION PLANS: The saviour

Heat action plans are a strategic framework developed to mitigate the growing risks of heatwaves due to climate change. According to the IMDs criteria, the plan state heatwaves as an increase of 4.5°C to 6.4°C above normal temperatures, with severe heatwaves exceeding 6.4°C. It identifies the most vulnerable regions, categorized as high risk areas. The plan outlines early warning systems, public health preparedness, and infrastructure adaptation to combat extreme temperatures. It emphasizes reserving hospital beds for heatstroke patients, ensuring water and electricity availability, and setting up cooling centers in urban and rural areas.

The plan also focuses on community engagement through media awareness campaigns, mobile alerts, and collaborations with local government bodies to educate people on preventive measures. The governance framework ensures coordinated action among district collectors, health departments, municipal corporations, and emergency services to implement heat mitigation strategies effectively.

Overall, the Heat Action Plan is a proactive policy initiative, aligning scientific research with disaster management to enhance climate resilience and reduce the devastating impacts of extreme heat on population and infrastructure. Through data driven planning, health interventions, and community preparedness, the plan aims to protect vulnerable communities and strengthen the state's adaptive capacity against rising temperatures

Report :Assessment of EU Adaptation Investment

Needs published on January 2026 classifies

heat stress on human health as the only catastrophic risk

Yet the EU's annual adaptation finance dedicated to heat-related measures falls dramatically short of what

economic modelling shows is needed.

Statement - Europe
lost 200 000 people to
heat in 4 years yet
nearly all of them
were preventable

The study estimates total **EU-wide adaptation investment needs at €69 billion per year** (€1.2 trillion cumulative NPV to 2050, discounted at 3%). Of this, heat-related expenditure cuts across multiple sectors health (occupational heat stress, urban cooling), infrastructure (building retrofitting for thermal resilience), food (livestock heat stress, crop adaptation), and ecosystems (wildfire prevention driven by heat-drought interaction). The current supply of adaptation finance from both public and private sources remains critically insufficient.

Hotspots and Regional Differentiation

Regions	Risk	Risk Index
Northern	Rising wildfire risk, changing fisheries, some urban heat	Limited
Southern	Heat stress, drought, wildfire, energy disruption, crop failure	Critical
Eastern and Central	Crop production, water scarcity, heat stress on agriculture & labour	Substantial
Western	Urban heat islands, infrastructure heat damage, flood-heat interaction	Substantial

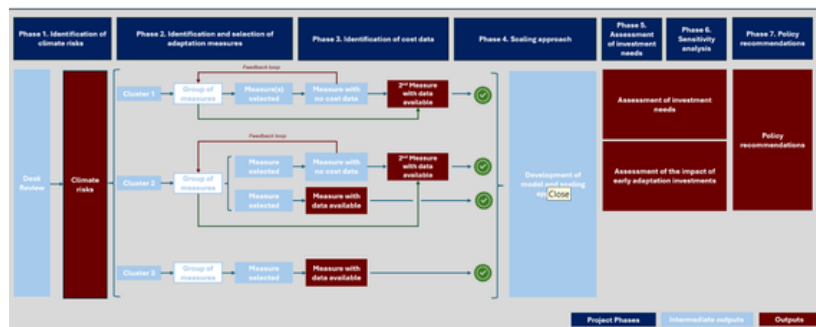
Notably, even Northern countries face heat-related economic disruption. Finland, Sweden, Latvia, and Germany are experiencing growing wildfire risks—a direct consequence of heat-drought synergies—that threaten carbon sinks and forest-based industries.

Extreme heat impacts economic output through five primary transmission channels, each with measurable fiscal and macroeconomic consequences:

#	Channel	Mechanism	Economic Impact
1	Labour Productivity	Heat stress reduces outdoor & physical worker output; occupational health & safety rules mandate work stoppages	€2.4 bn/yr in OHS adaptation; unreported GDP losses from reduced working hours in Southern Europe
2	Agricultural Output	Heat + drought reduces crop yields, increases livestock mortality; alters growing regions	€11.5 bn/yr food adaptation; €135 bn NPV in climate-smart agriculture alone
3	Energy Systems	Peak cooling demand spikes + reduced hydropower (heat-drought); transmission infrastructure stress	Southern Europe energy disruption classified as 'critical'; major investment trigger in grid adaptation

4	Built Environment	Thermal stress on infrastructure; urban heat island effect magnifies health costs; building overheating	€6.6 bn/yr in retrofitting (adaptation share); €120 bn NPV; only 10% of retrofit cost is pure adaptation
5	Public Health & Fiscal Costs	Heat-related mortality & morbidity increase healthcare expenditure; stress health system capacity	€7 bn/yr health sector adaptation (€118 bn NPV); Italy alone faces €714 mn/yr in wastewater upgrades linked to heat

Overview



the Policy

The EU's response to heat stress is fitted within the **European Green Deal** and the **2021 EU Climate Adaptation Strategy**, which calls for 'smarter, faster, and more systemic' adaptation across all Member States. The EUCRA, published as the primary risk reference classifies heat stress as the key systemic risk requiring urgent cross-border coordination. National responses are codified in National Adaptation Plans (NAPs) and National Adaptation Strategies (NAS), though the 2026 assessment found that most lack quantified, time-bound targets.

The 2026 assessment lists 55 adaptation measures across five clusters. The following are the primary heat-relevant interventions, along with their estimated costs:

- **Health Sector (Total: €7 bn/yr | €118 bn NPV)**

shaded rest areas, protective equipment for outdoor/manual workers, Green/hybrid urban interventions (parks, green roofs, reflective surfaces) to reduce urban heat islands, Expanded hospital capacity; heat-related illness protocols; emergency response planning ,Emergency medical preparedness; evacuation; protective equipment; post-fire public health

- **Infrastructure Sector (Total: €29 bn/yr | €521 bn NPV)**

Thermal insulation, passive cooling, shading, Heat-resistant rail joints, road surfaces, bridges; cooling stations at transport hubs, Adaptation of heat-sensitive historic structures; microclimate management, Thermal insulation, passive cooling, shading , cooling stations at transport hubs Adaptation of heat-sensitive historic structures; microclimate management

- **Food Sector (Total: €11.5 bn/yr | €200 bn NPV)**

Heat + drought tolerant varieties; precision irrigation; soil water retention
Climate-controlled barns; heat-tolerant breed development; adjusted feeding regimes
Breeding programmes to reduce heat stress and disease risks; genetic adaptation.

- **Ecosystems Sector (Total: €21 bn/yr | €366 bn NPV)**

Heat-driven fire risk; Mediterranean basin and Northern EU forests at expanding risk
Heat degrades topsoil; restoration maintains carbon sink & food security
Warmer temperatures, milder winters enable non-native species to thrive; threatening biodiversity and agriculture

While absolute investment needs are largest in Germany, France, Italy, and Spain, the relative fiscal burden falls most heavily on smaller Member States. The GDP-normalised comparison reveals a highly unequal distribution:

Table 8 Proportion of total adaptation investment needs against Member States' GDP.

	Ecosystems	Infrastructure	Health	Food	Economy and Finance	Cross-cutting measures	Total
EU 27	0.167%	0.190%	0.043%	0.051%	0.003%	0.003%	0.459%
Belgium	0.029%	0.134%	0.015%	0.042%	0.004%	0.003%	0.228%
Bulgaria	0.239%	0.093%	0.071%	0.038%	0.003%	0.004%	0.449%
Czechia	0.102%	0.154%	0.030%	0.033%	0.000%	0.004%	0.323%
Denmark	0.189%	0.107%	0.022%	0.073%	0.002%	0.001%	0.394%
Germany	0.049%	0.099%	0.015%	0.028%	0.002%	0.001%	0.193%
Estonia	0.614%	0.524%	0.098%	0.048%	0.003%	0.006%	1.296%
Ireland	0.073%	0.072%	0.018%	0.055%	0.002%	0.000%	0.220%
Greece	0.344%	0.180%	0.060%	0.108%	0.000%	0.003%	0.699%
Spain	0.152%	0.089%	0.037%	0.081%	0.003%	0.001%	0.363%
France	0.051%	0.175%	0.025%	0.063%	0.004%	0.001%	0.320%
Croatia	0.205%	0.473%	0.096%	0.051%	0.007%	0.003%	0.839%
Italy	0.083%	0.182%	0.055%	0.072%	0.006%	0.001%	0.400%
Cyprus	0.083%	0.114%	0.030%	0.041%	0.001%	0.002%	0.270%
Latvia	0.467%	0.776%	0.078%	0.052%	0.005%	0.006%	1.388%
Lithuania	0.278%	0.104%	0.034%	0.071%	0.002%	0.004%	0.495%
Luxembourg	0.019%	0.078%	0.006%	0.014%	0.002%	0.000%	0.120%
Hungary	0.165%	0.242%	0.046%	0.056%	0.000%	0.005%	0.516%
Malta	0.006%	0.099%	0.042%	0.010%	0.001%	0.001%	0.158%
Netherlands	0.023%	0.094%	0.015%	0.054%	0.000%	0.001%	0.187%
Austria	0.078%	0.138%	0.019%	0.039%	0.004%	0.001%	0.280%
Poland	0.114%	0.116%	0.033%	0.047%	0.004%	0.003%	0.318%
Portugal	0.163%	0.154%	0.049%	0.130%	0.003%	0.003%	0.502%
Romania	0.184%	0.137%	0.060%	0.055%	0.019%	0.003%	0.460%
Slovenia	0.053%	0.203%	0.038%	0.042%	0.007%	0.003%	0.347%
Slovakia	0.138%	0.277%	0.032%	0.028%	0.000%	0.007%	0.483%
Finland	0.384%	0.141%	0.074%	0.034%	0.003%	0.002%	0.637%

¹⁶ The net present value of GDP to 2050 is estimated based on the GDP growth projection of the European Commission's 2024 Ageing Report ([Underlying Assumptions and Projection Methodologies](#)).

The **2026 European Commission assessment** delivers a clear verdict: the EU's adaptation investment gap is large, structurally entrenched, and growing.

At €69 billion per year in estimated needs against an annual public supply of ~€20–33 billion (broad climate-relevant spending), the financing gap for heat adaptation alone runs into tens of billions annually. The economics are clear: adaptation delivers returns that exceed costs, particularly for no and low regret measures that generate benefits across mitigation, health, and ecosystem policy goals.

Extreme heat sits at the centre of this challenge. As the sole catastrophic risk in the 36-category EUCRA framework, heat stress is not merely a health issue, it is a labour market shock, a food security threat, an infrastructure liability, and an insurance market disruption. Its economic legs reach every productive sector of the European economy.

Observing HAP Economics from China's Lens

North China is set to experience its first major heat wave of 2026, with temperatures expected to reach 37°C in some areas.

North China set for first heat wave of 2026 in coming days

Source: Xinhua Editor: huaxia 2026-05-09 20:59:15



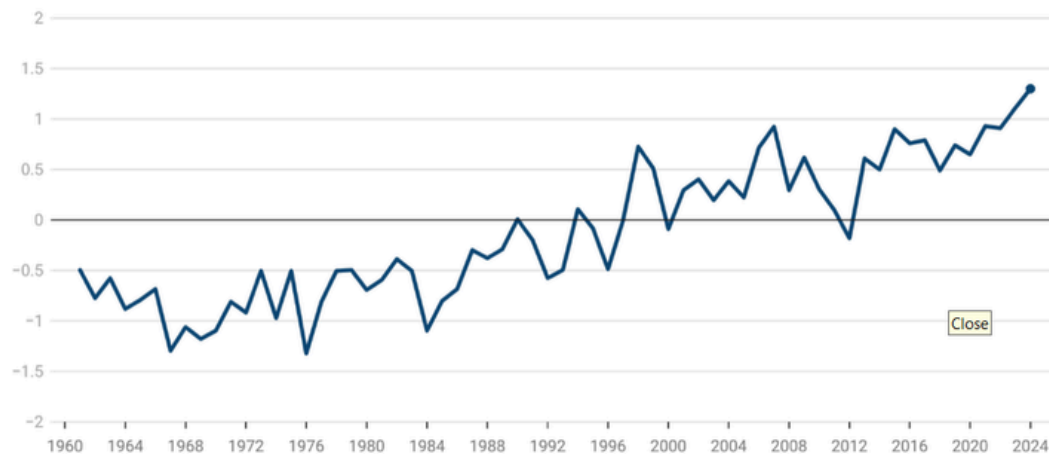
According to China's meteorological

authorities, rainfall will remain below normal over the next 10 days, while temperatures will stay 1–3°C above average, and up to 4–5°C above average in some locations.

the heat wave is expected to spread across major provinces and cities including Beijing, Tianjin, Hebei, Henan, Shanxi, Shandong, and Jiangsu. Beijing alone is forecast to see temperatures rise to 31–34°C early in the week.

A notable feature of this event is that North China is warming faster than southern regions, reversing the usual seasonal pattern where summer advances northward from the Yangtze River basin. Parts of the North China Plain may enter summer conditions earlier than southern China.

China annual average temperature (1961–2024), relative to the 1981–2010 average, degrees C

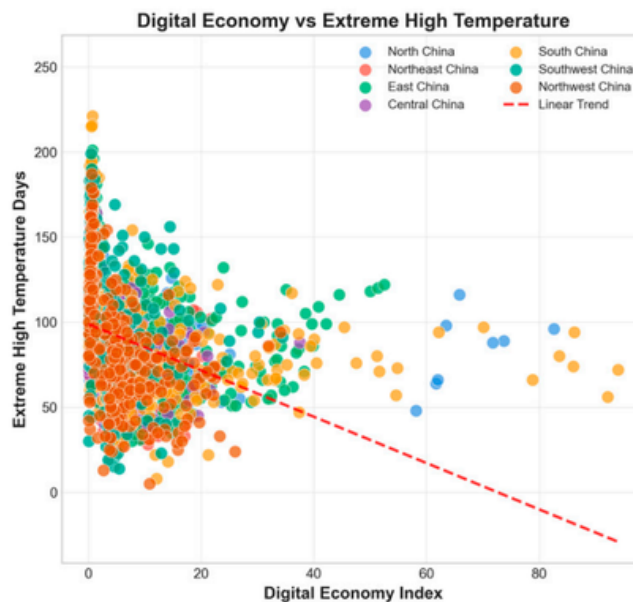


Source: China Meteorological Administration

CarbonBrief
CLEAR ON CLIMATE

Economic Insights: Extreme Heat & Economy in China

- **Short Term effects** :- Extreme heat has increased demand for online services (remote work, online education, e-commerce, delivery apps). Labor productivity loss has impacted economy a lot. Outdoor and non-air conditioned workers face reduced efficiency, lowering traditional economic output. Cooling costs: Data centers and IT infrastructure face sharply higher cooling expenses, crowding out energy use for other sectors.
- **Long Term effects** :- High cooling demand forced cross-regional power reallocation, reducing energy available for digital infrastructure in neighboring cities. Industrial chain disruption: Reduced production efficiency delays supply chains, spreading negative effects across regions, leading to industrial chain disruption.



China's National Climate Change : A Comprehensive Review , Targets, and Implementation Pathways Adaptation Strategy 2035

China's warming has outpaced this global average, with temperatures increasing at 0.26°C per decade between 1951 and 2020. Warming is not uniform across the country , Northeast China in particular has warmed faster than the national average, while North China has experienced significant warming combined with drying trends.

the strategy identifies substantial climate risks flowing into economic and social systems. These include disruption to agricultural production and crop layouts; increased meteorological hazards and pest pressures; growing health risks from extreme heat and allergic conditions; reduced safety and reliability of transport and energy infrastructure; and threats to urban lifeline systems. Longer term, the strategy warns of potential systemic financial and economic risks as climate change reshapes resource utilisation patterns and supply chains.

The strategy establishes following quantified and qualitative targets:

By 2030: Climate change observation, prediction, and risk-management systems basically complete; climate vulnerability of natural and socio economic systems significantly reduced; adaptation technology standards preliminarily established; phased results achieved in building a climate-adaptive society.

By 2035: Monitoring and early warning capacity reaches international advanced levels; climate risk management system mature; forest coverage 26%; wetland protection rate 60%; grassland vegetation coverage 60%; coastal wetland remediation reaches 50,000 hectares; shoreline restoration reaches 1,200 km; offshore high-quality water proportion fundamentally improved; climate-adaptive society construction basically complete.

These targets are planned to accomplish by developing sectors of Water resources, terrestrial Ecosystems, Marine and Coastal zones, Agriculture and Food security, Health and Public Sanitation, Infrastructure and Urban Environments

Vulnerable Areas : Five major strategic regions receive particularly detailed attention given the concentration of population, economic activity, and interconnected climate risks:

Beijing-Tianjin-Hebei Region: Targeted to lead national scientific research on climate adaptation, with international standard livable city clusters to be basically complete by 2035. Strict water resource management and the strictest total-consumption red lines are mandated.

Yangtze River Economic Belt: Prioritises ecosystem restoration, returning farmland to forests and wetlands, restoring aquatic biodiversity , alongside unified basin-wide water resource management and deepened joint scheduling of water projects.

Guangdong-Hong Kong-Macao Greater Bay Area: Building cross-boundary collaborative monitoring and emergency response for sea level rise, typhoons, and marine disasters, alongside urban green space expansion to combat heat-island effects.

Yangtze River Delta Integration Region: Focuses on joint ecological monitoring, inter-governmental compensation mechanisms, and construction of the Huangpu River estuary tidal-barrier project to address sea level and salt tide risks.

Yellow River Basin: Comprehensive water conservation, integrated protection of mountains-rivers-forests farmlands lakes-grasslands-deserts, and anti-poverty efforts in ecologically fragile upper reaches.

Implementation

Governance Structure: China's climate adaptation efforts will be coordinated by the Ministry of Ecology and Environment, which will oversee implementation and guide regional adaptation planning. All ministries must prepare sector-specific adaptation plans, while provincial governments will develop local plans based on regional climate risks. A comprehensive monitoring and evaluation system will regularly assess policy effectiveness, supervise implementation, and address emerging challenges.

Fiscal and Financial Support : The strategy sets out a multi-layered financial architecture. Carbon-emission reduction support tools are to guide financial institutions toward green funding. A step-by-step mandatory disclosure system for climate and environmental information will cover all financial institutions and financing entities. Climate-risk stress tests are to be promoted for banking, securities, and insurance industries. By 2035, the early warning mechanism for climate-change related financial risks is to be fully operational, with significantly enhanced risk-identification capabilities across financial institutions.

Innovative financial products ,sustainability-linked bonds, catastrophe insurance, and climate-risk insurance are to be developed in key sectors. International funds, bilateral and multilateral cooperation loans, and performance fund mechanisms such as the Green Climate Fund and Global Environment Facility are to be actively courted

Conclusion :

China's NCCAS 2035 represents a comprehensive, quantitatively grounded national commitment to managing the risks of an already-changing climate. The strategy is notable for its integration of sector-specific numerical targets (water caps, land coverage rates, infrastructure standards, coastal restoration quotas) with regional differentiation that reflects China's vast geographic diversity, and for its explicit framing of adaptation as inseparable from rather than secondary to the country's mitigation and development goals.

The strategy's credibility rests on the ambition and specificity of its targets, the breadth of its governance architecture, and the mobilisation of a multi-layered financial system to back adaptation investment. Key challenges acknowledged by the strategy itself include the insufficient current depth of climate impact assessment, the need to more fully mainstream adaptation into local government priorities, and the gap between existing adaptation measures and the quality of resilience required for a 'Big Beautiful China' by 2035.

Achieving the 2035 vision – an internationally advanced monitoring system, a mature climate-risk management framework, well-established technology and standards systems, and a basically complete climate-adaptive society – will require sustained, coordinated action across all levels of government, industry, finance, and civil society over the next decade.

